

Eroze Barua

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EDUCATION

UPES Dehradun

B.Tech in Computer Science (AI & ML)

CGPA: 7.9 May 2026

INTERNSHIP

IBM

Jun 2025 – Aug 2025

AI & ML Engineering Intern

- Built a clinical trial matching system from scratch over 370K+ records, letting patients and researchers find relevant trials in seconds instead of manually scanning databases.
- Designed the retrieval and ranking pipeline (Precision@10: 0.45, NDCG@10: 0.46) using Gemini 1.5 Pro to classify patient conditions across 14 medical categories with 93.57% accuracy.
- Reduced end-to-end response latency by 28% by optimising inference and backend workflows; shipped with Django, secure user auth, and a clean frontend.

PROJECTS

AI Learning Resource Explorer | *Django, FastAPI, PostgreSQL, Redis, Docker Compose*

- Built a dual-service backend for discovering learning resources, splitting auth/CRUD (Django) from recommendations (FastAPI) behind a shared PostgreSQL database.
- Implemented Postgres full-text search with ranking, plus a Redis caching layer with live HIT/MISS visibility on both search and dashboard stats.
- Deployed as a multi-container Docker Compose stack with JWT-based auth and full CRUD across resources, categories, tags, and user interactions.

FlexiBill Sentinel | *FastAPI, PostgreSQL, LangGraph, Gemini API, Docker Compose*

- Built an AI billing oversight system with a LangGraph agent pipeline to classify and flag anomalous transactions end-to-end.
- Designed a pooled PostgreSQL schema behind a FastAPI service layer, decoupling agent logic from data access.
- Deployed live via Docker Compose across independently scalable API, database, and agent services.

RTO Sentinel | *FastAPI, Django, Scikit-Learn, Docker Compose*

- Built an order risk-scoring system to flag high-risk e-commerce orders before dispatch, in response to Razorpay's open problem statement.
- Achieved 87.2% accuracy, 0.92 precision, 0.85 recall on the fraud class using 8+ engineered features.
- Deployed as a multi-service Docker Compose stack with Pydantic-validated inference endpoints.

TECHNICAL SKILLS

Languages: Python, C++

AI / ML: PyTorch, Scikit-Learn, Transformers, LLMs, LangChain, LangGraph, RAG, NLP, HuggingFace, Ollama

Backend: FastAPI, Django, Pydantic, Redis

Deployment: Docker, Docker Compose

Data: PostgreSQL, SQLite, Pandas, NumPy, Feature Engineering

Tools: Git

Core CS: Data Structures & Algorithms, OS, DBMS, Computer Networks, OOP

ACHIEVEMENTS & CERTIFICATIONS

- Ranked 26/111 teams in Kaggle Multi-Instance Object Detection Challenge.
- **Oracle Cloud Infrastructure Generative AI Professional** (2025)
- **Oracle Cloud Infrastructure Data Science Professional** (2025)